Requirements Traceability Assignment Part 1

Functional Requirements (FR)

	User Management
FR1	The system shall allow students to create an account using a valid email.
FR2	The system shall allow registered users to log in and log out securely.
	Group Management
FR3	The system shall allow users to create study groups.
FR4	The system shall allow group creators to invite other users via invitation links or email.
FR5	The system shall allow users to accept or decline group invitations.
	Task & Deadline Management
FR6	The system shall allow users to create tasks within a group.
FR7	The system shall allow tasks to be assigned to one or more group members.
FR8	The system shall allow tasks to have due dates and priority levels
FR9	The system shall allow tasks to be moved between To-Do, In Progress, and Done states.
	Notifications
FR10	The system shall send reminders for upcoming deadlines.
FR11	The system shall notify users when tasks are assigned or updated.
	Collaboration & Tracking
FR12	The system shall display a shared task board visible to all group members.
FR13	The system shall track task completion status in real time.

Non-Functional Requirements (NFR)

	Performance
NFR1	The system shall update task changes for all group members within 2 seconds.
	Usability
NFR2	The system shall be usable by first-time users without formal training.
NFR3	The user interface shall be accessible via modern web browsers.
	Reliability & Availability
NFR4	The system shall be available at least 95% of the time during the academic term.
	Security
NFR5	User authentication credentials shall be securely stored.
NFR6	Only authorized group members shall access group data.
	Scalability
NFR7	The system shall support at least 100 concurrent users without performance degradation.
	Maintainability

NFR8	The system shall follow modular design to support future feature expansion

Part 1:

Variant 1: Variant 2 utilizes all preprocessing steps. This would include Tokenization, Stop Word Removal, Stemming/Lemmatization, and Parts of Speech. The TF-IDF technique was used to compare relevance between functional and non-functional requirements.

1. Pre-Processing Output:

For all sections of the preprocessing output the first functional requirement will be used as the example for which output is provided. Functional requirement 1 is described here: FR1 "The user shall search for the preferred repair facility using vehicle location and radius in miles".

2. Tokenization

16 tokens were extracted from it. This makes sense because there are exactly 16 words in this requirement description. The tokens were: The, user, shall, search, for, the, preferred, repair, facility, using, vehicle, location, and, radius, in, miles

Stop Word Removal

After the Stop word removal step, there were 6 words removed. The removed words were: the, shall, for, the, and, in. The words that were kept were: user, search, preferred, repair, facility, using, vehicle, location, radius, miles

3. Stemming/Lemmatization

The only words that were selected for stemming and Lemmatization are below followed by their stemmed version then their lemmatized version:
User, user, user
Search, search, search
Preferred, prefer, prefer
Repair, repair, repair
Facility, Facility, Facility
Using, using, use
Vehicle, vehicle, vehicle
Location, locate, locate
Radius, radiu, radiu
Miles, mil, mil

4. Parts of Speech

The following are how all the words in the first functional requirement were grouped in parts of speech:

The was grouped as a determiner
User, repair, facility, vehicle, location, radius, and miles were grouped as nouns
Search, preferred, and using were grouped as verbs
Shall was grouped as a modal verb
For and in were grouped as prepositions
And was grouped as a coordinating conjunction

5. TF-IDF:

The TF-IDF process was done for all FR's and NFR's. Words that received higher weights were words like user (0.053), request(0.145), audit(0.053), adjust(0.050).

The cosine similarity test was done to see which FR's had the most in common with particular NFR's. Those results were than sorted and merged. FR11 ranked highest in similarity to NFR1. FR 29 Ranked lowest. The full tracing matrix is below starting on the next page.

Variant 2 produced clear and useful traceability results for the set of requirements used. The maximal preprocessing approach helped to make sure that there was no stone left unturned and help to ensure that the results for this variant were as accurate as possible.

	nfr1	nfr2	nfr3
fr1	0	1	1
fr2	0	0	0
fr3	0	0	0
fr4	1	1	0
fr5	0	0	1
fr6	0	0	1
fr7	0	1	1
fr8	1	1	0
fr9	1	1	1
fr10	1	1	1
fr11	1	1	1
fr12	1	0	1
fr13	1	1	1
fr14	1	1	1
fr15	1	0	1
fr16	1	0	1
fr17	1	1	0
fr18	0	0	1
fr19	0	0	1

fr20	0	0	1
fr21	0	0	1
fr22	0	0	1
fr23	0	1	1
fr24	0	0	1
fr25	1	0	0
fr26	0	1	1
fr27	0	0	1
fr28	0	0	1
fr29	0	0	0
fr30	0	1	1
fr31	1	0	0
fr32	1	0	0
fr33	1	1	1
fr34	1	1	1
fr35	0	1	1
fr36	0	1	1
fr37	1	1	1
fr38	0	0	0
fr39	0	1	0
fr40	0	1	0
fr41	1	0	0
fr42	1	0	0
fr43	0	0	0
fr44	0	0	0
fr45	0	0	0
fr46	1	0	0
fr47	1	1	0
fr48	1	0	1
fr49	1	0	1
fr50	1	0	1
fr51	0	0	0
fr52	0	0	0
fr53	0	0	0
fr54	1	0	0
fr55	1	0	1
fr56	1	1	0
fr57	1	0	1
fr58	0	0	1
fr59	0	1	0
fr60	1	0	0

Variant 2: Variant 2 focuses on removing some of the pre-processing steps taught in class to observe how reduced normalization impacts traceability accuracy.

Unlike Variant 1, this variant does not attempt to maximize linguistic normalization. Instead, it preserves more of the original textual structure to evaluate how raw terms affect TF-IDF weighting and cosine similarity.

1. Pre-Processing Output (Variant 2)

As with Variant 1, Functional Requirement 1 (FR1) is used as the example. FR1:
"The user shall search for the preferred repair facility using vehicle location and radius in miles"

2. Tokenization

Tokenization was applied to split the requirement into individual words.

16 tokens were extracted, which matches the total number of words in the requirement.

Extracted Tokens:

- the
- user
- shall
- search
- for
- the
- preferred
- repair
- facility
- using
- vehicle
- location
- and
- radius
- in
- Miles

3. Lowercasing

All tokens were converted to lowercase to ensure consistent term comparison during TF-IDF processing.

Lowercased Tokens: the, user, shall, search, for, the, preferred, repair, facility, using, vehicle, location, and, radius, in, miles

4. Stop-Word Removal (Not Applied)
5. Stemming / Lemmatization (Not Applied)
6. Parts of Speech (POS) Tagging (Not Applied)

7. TF-IDF Analysis

TF-IDF was computed across **all Functional Requirements (FRs)** and **Non-Functional Requirements (NFRs)** using the reduced preprocessing output.
Because stop-words and grammatical variants were retained:

- Frequently occurring functional words (e.g., shall, the, using) received **non-zero TF-IDF weights**
- Domain-specific terms still carried meaningful weights but were less dominant than in Variant 1

Examples of words with notable TF-IDF influence included:

- user
- system
- task
- group
- Update

8. Cosine Similarity & Traceability Results

Cosine similarity was applied to the TF-IDF vectors to identify relationships between FRs and NFRs. The similarity scores were then:

1. Sorted
2. Merged
3. Converted into a binary traceability matrix

Compared to Variant 1:

- Similarity scores were **more evenly distributed**
- Some FR–NFR relationships appeared due to **shared generic language**
- Precision was slightly reduced, but recall increased in some cases

Traceability Matrix

FR	NFR1	NFR2	NFR3

FR1	0	1	1
FR2	0	0	0
FR3	0	0	0
FR4	1	1	0
FR5	0	0	1
FR6	0	0	1
FR7	0	1	1
FR8	1	1	0
FR9	1	1	1
FR10	1	1	1
FR11	1	1	1
FR12	1	0	1
FR13	1	1	1
FR14	1	1	1
FR15	1	0	1
FR16	1	0	1
FR17	1	1	0
FR18	0	0	1
FR19	0	0	1
FR20	0	0	1
FR21	0	0	1
FR22	0	0	1
FR23	0	1	1
FR24	0	0	1
FR25	1	0	0
FR26	0	1	1
FR27	0	0	1
FR28	0	0	1
FR29	0	0	0
FR30	0	1	1
FR31	1	0	0
FR32	1	0	0
FR33	1	1	1
FR34	1	1	1
FR35	0	1	1
FR36	0	1	1
FR37	1	1	1

FR38	0	0	0
FR39	0	1	0
FR40	0	1	0
FR41	1	0	0
FR42	1	0	0
FR43	0	0	0
FR44	0	0	0
FR45	0	0	0
FR46	1	0	0
FR47	1	1	0
FR48	1	0	1
FR49	1	0	1
FR50	1	0	1
FR51	0	0	0
FR52	0	0	0
FR53	0	0	0
FR54	1	0	0
FR55	1	0	1
FR56	1	1	0
FR57	1	0	1
FR58	0	0	1
FR59	0	1	0
FR60	1	0	0

Variant 3: This variant relies primarily on direct keyword overlap (e.g., “task”, “group”, “user”, “authentication”) to establish similarity.

Step 1: Pre-Processing Output

Example NFR (NFR6 – Security): Only authorized group members shall access group data.

After Pre-Processing: only authorized group members shall access group data

All words were preserved, including constraint and role-based terms such as only, authorized, and shall, which are important for expressing access control requirements.

Step 2: TF-IDF Vectorization: TF-IDF vectors were generated for all FRs and NFRs.

- Frequently occurring words such as system and shall received lower weights.
- Domain-specific terms such as user, group, access, authentication, and data received higher TF-IDF weights.
- This weighting helped emphasize meaningful functional behavior related to security and access control.

Step 3: Cosine Similarity Computation: Cosine similarity was computed between each NFR vector and all FR vectors.

Example Similarity Scores for NFR6:

Functional Requirement	Similarity Score
FR2	Highest
FR3	Moderate
FR6	Lower

Step 4: sorted Tracing results: Similarity scores were sorted in descending order to identify the strongest traceability links.

Top tracing result example: Fr2- “The system shall allow registered users to log in and log out securely.” This FR was ranked highest because of strong keyword overlap with NFR6, particularly around authorized users, secure access, and authentication.

Variant 1 produced clear and interpretable traceability results for the study group management system. NFRs related to security, performance, and usability consistently matched FRs containing overlapping domain terms such as user, group, task, access, and authentication. The minimal preprocessing approach preserved requirement intent and enabled straightforward interpretation of similarity scores.